

Name: _____

Solving Radical Equations

On-Level Notes

Day 1: Solving When the Radical is Already Isolated

Process

Step 1: Make sure the radical is alone on one side.**Step 2:** Square both sides of the equation.**Step 3:** Solve for x .**Step 4:** Check your answer in the original equation.**Example 1** Solve the equation. Check your solution.

A. $\sqrt{x + 3} = 5$

B. $\sqrt{2x - 1} = 7$

C. $\sqrt{4x + 5} = 9$

Connection to Rational Exponents: Remember that $\sqrt{x} = x^{1/2}$. This means radical equations can also be written using rational (fractional) exponents!

Examples: $\sqrt{x+3} = x^{1/2} + 3$ and $x^{1/2} = 5$ is the same as $\sqrt{x} = 5$

To solve equations with rational exponents, use the SAME PROCESS!

Process for Rational Exponents

Step 1: Make sure the term with the rational exponent is alone on one side. **Step 2:** Raise both sides to the reciprocal power (this eliminates the exponent).

- If you have $x^{1/2}$, raise both sides to the 2nd power
- If you have $x^{1/3}$, raise both sides to the 3rd power

Step 3: Solve for x . **Step 4:** Check your answer in the original equation.

Example 2 Solve the equation. Check your solution.

A. $(x - 4)^{1/2} = 6$

B. $(3x + 5)^{1/2} = 8$

Day 2: Isolating the Radical First & No Solution Cases

Process

Step 1: Isolate the radical (get it alone on one side).

Step 2: Check – if the radical equals a NEGATIVE number, there is NO SOLUTION.

Step 3: Square both sides of the equation.

Step 4: Solve for x .

Step 5: Check your answer in the original equation.

Key Concept: A square root can NEVER equal a negative number. If you isolate the radical and it equals a negative, STOP – there is no solution!

Example 2 Solve the equation. Check for extraneous solutions.

A. $\sqrt{x+5} + 3 = 7$

C. $\sqrt{x+1} + 6 = 2$

B. $\sqrt{2x-3} - 4 = 2$

D. $3(x-4)^{\frac{1}{2}} = 15$

Day 3: Equations with Two Radicals & Applications

Process (Two Radicals – One on Each Side)**Step 1:** Make sure you have one radical on each side of the equation.**Step 2:** Square both sides – the radicals will disappear!**Step 3:** Solve the linear equation.**Step 4:** Check your answer in the original equation.**Example 3** Solve the equation. Check for extraneous solutions.

A. $\sqrt{3x+1} = \sqrt{x+9}$

B. $\sqrt{5x-2} = \sqrt{2x+7}$

C. $\sqrt{2x+10} = (5x+16)^{\frac{1}{2}}$

Radical Equations Application

Example 4 The time t (in seconds) it takes an object to fall from a height h (in feet) is given by the formula:

$$t = \sqrt{\frac{h}{16}}$$

If an object takes 2.5 seconds to hit the ground, from what height was it dropped?

Example 5: Distance to Horizon: The distance d (in miles) that you can see to the horizon from a height h (in feet) above the ground is given by:

$$d = \sqrt{1.5h}$$

If you can see 15 miles to the horizon, how high above the ground are you?